VPC Peering Project: Resource Inventory

EWB Cloud Labs / AWS Architect Associate / Fill in the blanks as you build

1. Student details

Field	Value
Name	Jiya Mary Jacob
Batch	AWS Internship Batch 1
Project title	AWS VPC Peering
Region	ap-south-1 (Mumbai)
Availability Zone used	ap-south-1a
Start date	28 September 2026
Completion date	21st October 2026
Tags on every resource	Project = EWB-VPC-Peering, Student =Jiya

2. VPCs

Name tag	VPC ID	IPv4 CIDR	IPv6	Tenancy	Tags added
ewb-vpc-a	vpc-074f44afffd33e8b0	10.10.0.0/16	None	Default	Project = EWB-VPC-Peering, Student = Jiya
ewb-vpc-b	vpc-071028f353ae50f21	10.20.0.0/16	None	Default	Project = EWB-VPC-Peering, Student = Jiya

3. Subnets

Name tag	Subnet ID	VPC	CIDR	AZ	Auto-assign public IPv4	Tags added
ewb-a-public	subnet-00a5804e737fb0429	ewb-vpc-a	10.10.1.0/24	aps1-az1 (ap-south-1a)	Off (public IP set on bastion at launch)	Project = EWB-VPC-Peering, Student = Jiya

Name tag	Subnet ID	VPC	CIDR	AZ	Auto-assign public IPv4	Tags added
ewb-a-private	subnet-0234c35664492b902	ewb-vpc-a	10.10.2.0/24	aps1-az1 (ap-south-1a)	Off	Project = EWB-VPC-Peering, Student = Jiya
ewb-b-private	subnet-035b918b997acd8f7	ewb-vpc-b	10.20.1.0/24	aps1-az1 (ap-south-1a)	Off	Project = EWB-VPC-Peering, Student = Jiya

4. Internet gateway

Name tag	IGW ID	Attached to	State	Tags added
ewb-igw-a	igw-07c4520711c1268fa	ewb-vpc-a	Attached	Project = EWB-VPC-Peering, Student = Jiya

VPC B has no internet gateway.

5. Route tables

5.1 Table IDs and associations

Name tag	Route table ID	VPC	Explicitly associated subnet	Tags added
(main table of VPC A, left unchanged)	rtb-079ec7b825df805f8	ewb-vpc-a	None	
ewb-rt-a-public	rtb-0211d8a5965a3eda4	ewb-vpc-a	ewb-a-public	Project = EWB-VPC-Peering, Student = Jiya
ewb-rt-a-private	rtb-0af2a2caab36a0585	ewb-vpc-a	ewb-a-private	Project = EWB-VPC-Peering, Student = Jiya
(main table of VPC B, left unchanged)	rtb-0549dcd7b622b66b9	ewb-vpc-b	None	

Name tag	Route table ID	VPC	Explicitly associated subnet	Tags added
ewb-rt-b-private	rtb-04039a2202e5eb420	ewb-vpc-b	ewb-b-private	Project = EWB-VPC-Peering, Student = Jiya

5.2 Routes: before peering (Checkpoint 01)

Route table	Destination	Target	Done
ewb-rt-a-public	10.10.0.0/16	local	Yes
ewb-rt-a-public	0.0.0.0/0	igw-07c4520711c1268fa	Yes
ewb-rt-a-private	10.10.0.0/16	local	Yes
ewb-rt-b-private	10.20.0.0/16	local	Yes

5.3 Routes: added after peering is Active (Checkpoint 04)

Route table	Destination	Target	Purpose	Done
ewb-rt-a-private	10.20.0.0/16	pcx-018895e855a398629	App-A to VPC B	Yes
ewb-rt-b-private	10.10.0.0/16	pcx-018895e855a398629	Return path to VPC A	Yes
ewb-rt-a-public	10.20.0.0/16	pcx-018895e855a398629	Bastion SSH to App-B	Yes

6. Security groups

6.1 Group IDs

Name	Security group ID	VPC	Description used	Tags added
ewb-sg-bastion	sg-012fdb869c6b1077f	ewb-vpc-a	SSH access to bastion from my IP	Project = EWB-VPC-Peering, Student = Jiya
ewb-sg-app-a	sg-0a28c745319e85766	ewb-vpc-a	Private App-A in VPC A, SSH from bastion and ping from App-B	Project = EWB-VPC-Peering, Student = Jiya

Name	Security group ID	VPC	Description used	Tags added
ewb-sg-app-b	sg-002f9b9e3f8015ec8	ewb-vpc-b	Private App-B in VPC B, SSH from bastion and ping from App-A	Project = EWB-VPC-Peering, Student = Jiya

6.2 Inbound rules

Security group	Type	Protocol / port	Source	Rule description	Done
ewb-sg-bastion	SSH	TCP 22	(my public IP)/32	SSH from my laptop	Yes
ewb-sg-app-a	SSH	TCP 22	10.10.1.10/32	SSH from bastion private IP	Yes
ewb-sg-app-a	Custom ICMP - IPv4	Echo request	10.20.1.10/32	ping from app B over peering	Yes
ewb-sg-app-b	SSH	TCP 22	10.10.1.10/32	SSH from bastion private IP	Yes
ewb-sg-app-b	Custom ICMP - IPv4	Echo request	10.10.2.10/32	ping from app A over peering	Yes

Outbound rules: default allow-all on all three groups. My public IP used for the bastion rule: 65.0.93.169

7. Key pair

Name	Type	Format	Local file location	Tags added
ewb-lab	RSA	.pem	~/.ssh/ewb-lab.pem	Project = EWB-VPC-Peering, Student = Jiya

8. EC2 instances (Checkpoint 02)

Name	Instance ID	AMI	Type	VPC / subnet	Private IP	Public IPv4	Security group	State / status checks
ewb-bastion	i-007009e53589bd038	Amazon Linux 2023	t3.micro	ewb-vpc-a / ewb-a-public	10.10.1.10		ewb-sg-bastion	
ewb-app-a	i-06031e6f18aa79a9f	Amazon Linux 2023	t3.micro	ewb-vpc-a / ewb-a-private	10.10.2.10	None	ewb-sg-app-a	
ewb-app-b	i-0c07959a5eb16b3be	Amazon Linux 2023	t3.micro	ewb-vpc-b / ewb-b-private	10.20.1.10	None	ewb-sg-app-b	

Root volume IDs (for cleanup):

bastion : vol-0afcdea04bad7d627

App-A : vol: 03efac937c3c63e15

App-B vol: 0426b221ab7fd62dd

9. Peering connection (Checkpoint 03)

Field	Value
Name	ewb-a-b-peer
Peering connection ID (pcx-...)	pcx-018895e855a398629
Requester VPC (ID / CIDR)	ewb-vpc-a / 10.10.0.0/16
Accepter VPC (ID / CIDR)	ewb-vpc-b / 10.20.0.0/16
Account / Region	My account / This Region
Status	Active

10. SSH setup

Item	Value
Bastion public IP used in ~/.ssh/config	65.0.93.169
chmod 400 done	Yes
ssh ewb-bastion works	Yes
ssh ewb-app-a works (before peering)	Yes
ssh ewb-app-b works (after routes)	Yes

11. Test log

Record real output only. The expected results are acceptance criteria.

Test	Where / command	Expected	Actual result (paste summary)	Pass
01 Baseline	App-A, before peering: ping -c 4 10.20.1.10	No replies	No replies from 10.20.1.10, 100% packet loss (no peering or routes yet)	Yes
02 Forward path	App-A, after routes: ping -c 4 10.20.1.10	4 replies, 0% loss	4 packets transmitted, 4 received, 0% packet loss	Yes
03 Reverse path	App-B, after routes: ping -c 4 10.10.2.10	4 replies, 0% loss	4 packets transmitted, 4 received, 0% packet loss	Yes
04 Private SSH	Laptop: ssh ewb-app-b	Shell on App-B via bastion	Shell opened on App-B (10.20.1.10) through the bastion using ProxyJump	Yes
05 Isolation	EC2 details for App-A and App-B	No public IPv4 on either	Public IPv4 address field empty on both App-A and App-B	Yes

Test	Where / command	Expected	Actual result (paste summary)	Pass
06 Controlled failure	ICMP rule removed from ewb-sg-app-b, ping from App-A	Replies stop	With the ICMP rule removed, ping from App-A got no replies (100% packet loss)	Yes
07 Recovery	Exact rule restored, ping again	Replies resume	After restoring the exact rule, 4 packets received, 0% packet loss	Yes

Also record the output of hostname and hostname -I on each private host:

App-A: ip-10-10-2-10.ap-south-1.compute.internal / 10.10.2.10

App-B: ip-10-20-1-10.ap-south-1.compute.internal/ 10.20.1.10

12. What I learned

VPC peering is only the link between two VPCs, and traffic uses it only when both sides have the right routes. I needed a route in each private route table (and in the public one for the bastion) pointing at the peering connection, because replies need a path back as well as the request. Security groups still decide what is allowed, and the controlled failure showed this: removing the ICMP Echo rule stopped the ping even though the peering and routes were fine. Using a bastion with ProxyJump let me reach both private servers without giving them public IPs or copying the key onto any instance. App-B sees the bastion's private IP as the SSH source, so that is the address its security group rule must allow. I also learned that peering is non-transitive and gives VPC B no internet access through VPC A, and that SSH is strict about file permissions on keys and config files.